Thermodynamic description of the Al–Bi system: a synthetic benchmark document

ORCHID synthetic benchmark (generated; no human authors)

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Abstract

This synthetic document describes the Al–Bi binary system as calculated with pycalphad from the SGTE dataset AlBi-06Kim.tdb; every figure, table and stated value is generated, and the exact values are kept in a separate truth key. The highest invariant reaction, L1 → (Al) + L2, lies at 658.6 °C.

1. Introduction

The Al–Bi system belongs to the 20 binaries of the ORCHID benchmark, chosen to cover diagram types such as miscibility gap, monotectic. The thermodynamic description is the SGTE collection entry [1], which reproduces the assessment of S.S. Kim, T.H. Sanders, Jr., Modelling Simul. Mater. Sci. Eng., 14, 1181-88(2006). [2]. The calculations use pycalphad 0.11.2 at 1 atm.

2. Phase diagram and invariant reactions

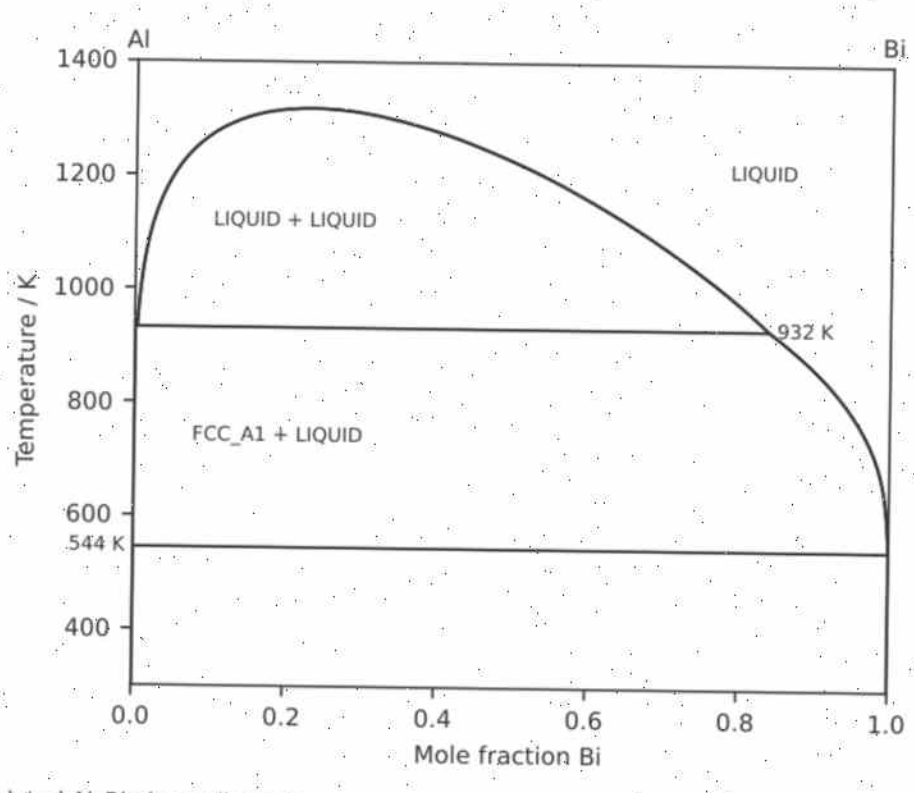


Fig. 1. Calculated Al–Bi phase diagram.

The monotectic reaction L1 → (Al) + L2 is calculated at 658.6 °C, with L1 at x(Bi) = 0.003. The dataset places a eutectic (L → (Al) + (Bi)) at 271.2 °C; the composition of L is x(Bi) = 0.999.

Table 1 compares the calculated temperatures with experimental values from the literature [3–5]; in this synthetic document those values and references are fictitious.

Table 1. Invariant reactions and congruent points, calculated and experimental (Al–Bi).

Reaction	Type	T / °C			Compositions (x(Bi))		
		this work	exp.[b]	Ref.	phase 1	phase 2	phase 3
L1 → (Al) + L2	monotectic	658.6	651.6	[3]	0.003	0.001	0.840
L → (Al) + (Bi)	eutectic	271.2	266.2	[3]	0.999	0.000	1.000

[b] Experimental temperatures are fictitious values attributed to invented references [3–5] for testing evidence-class extraction; they are not data.

3. Phase stability

Table 2. Solid phases: structure, stability range and homogeneity range (Al–Bi).

Phase	Pearson	SB	Prototype	T min (°C)	T max (°C)	Range (at.% Bi)	T (°C)
(Al)	cF4	A1	Cu	< 27	660.3	0.0–0.1	657
(Bi)	hR2	A7	α-As	< 27	271.4	100.0	27

Structure labels follow the TDB phase names (– where the name does not identify a structure). Stability range: temperatures between which the phase is stable in some part of the diagram.

The upper stability limit of (Bi) is 271.4 °C.

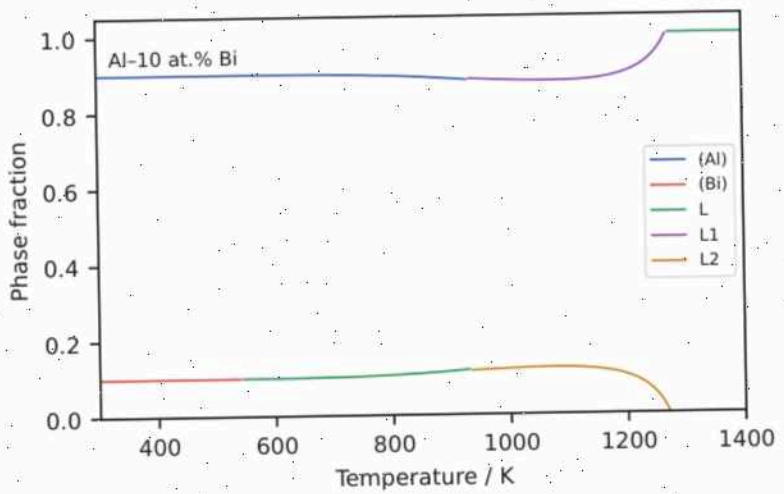


Fig. 2. Calculated equilibrium phase fractions versus temperature for Al–10 at.% Bi.

4. Thermodynamic properties

At 1400 K the integral mixing enthalpy of the liquid has its maximum, 19170 J/mol, at x(Bi) = 0.440. For the equiatomic liquid at 1400 K, the Gibbs energy of mixing is −7.71 kJ/mol-atom.

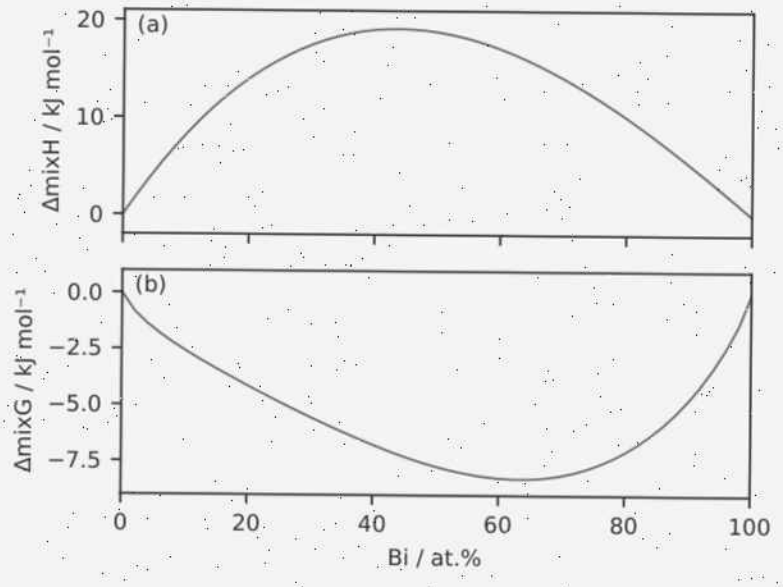


Fig. 3. Calculated (a) mixing enthalpy and (b) mixing Gibbs energy of the liquid at 1400 K.

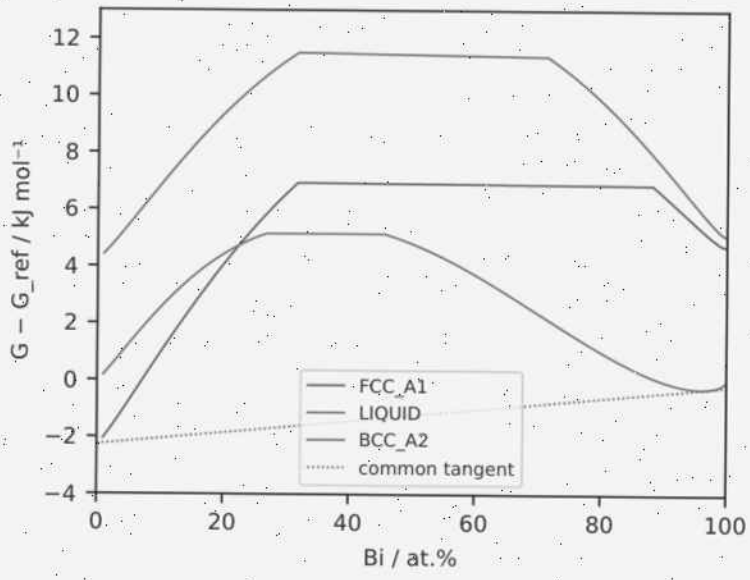


Fig. 4. Calculated Gibbs energy curves at 740 K.

Table 3. Redlich–Kister parameters of the liquid (Al–Bi).

	Liquid: L(ν) = a + b·T + c·T·ln T		
ν	**a**	**b**	**c**
L0	75351.7	−52.8	0.0
L1	20421.2	−3.7	0.0

Values as printed in the TDB file.

Table 4. Integral and partial mixing properties of the liquid at 1400 K (Al–Bi).

	Liquid at 1400 K					
x(Bi)	ΔmixH / J mol⁻¹	ΔmixG / J mol⁻¹	a(Al)	a(Bi)	ΔH(Al) / J mol⁻¹	ΔH(Bi) / J mol⁻¹
0.100	8252	−2558	0.9323	0.2087	1284	70960
0.300	17539	−5530	0.8750	0.2803	10090	34921

Table 4 (continued).

	Liquid at 1400 K					
x(Bi)	ΔmixH / J mol⁻¹	ΔmixG / J mol⁻¹	a(Al)	a(Bi)	ΔH(Al) / J mol⁻¹	ΔH(Bi) / J mol⁻¹
0.500	18838	−7710	0.7153	0.3717	23943	13733
0.700	14108	−8090	0.3623	0.5725	38924	3473
0.900	5311	−4753	0.0585	0.8709	51110	223

Reference states: pure liquid Al and Bi at 1400 K; per mole of atoms. ΔH: partial molar enthalpies of mixing.

5. Conclusions

This document is synthetic and exists only to test extraction tools; its truth key lists every boundary, field, invariant, plotted point, table cell and stated value.

References

[1] B. Hallstedt, SGTE binary dataset collection, Calphad 89 (2025) 102833 (CC BY 4.0).

[2] S.S. Kim, T.H. Sanders, Jr., Modelling Simul. Mater. Sci. Eng., 14, 1181-88(2006).

[3] K. Synthetica, L. Exemplar, J. Fict. Phase Equilib. 12 (1987) 345–352 — fictitious reference, invented for this synthetic benchmark.

[4] M. Placeholder, Z. Notreal, Fict. Thermochim. Acta 41 (1994) 77–89 — fictitious reference, invented for this synthetic benchmark.

[5] R. Imaginary, Proc. Synth. Calorim. Soc. 8 (2003) 1102–1110 — fictitious reference, invented for this synthetic benchmark.